Barış Yamuksuz 22202238

1.

s_1	s_2	s_3	f
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

2. Sum-of-Products expression:

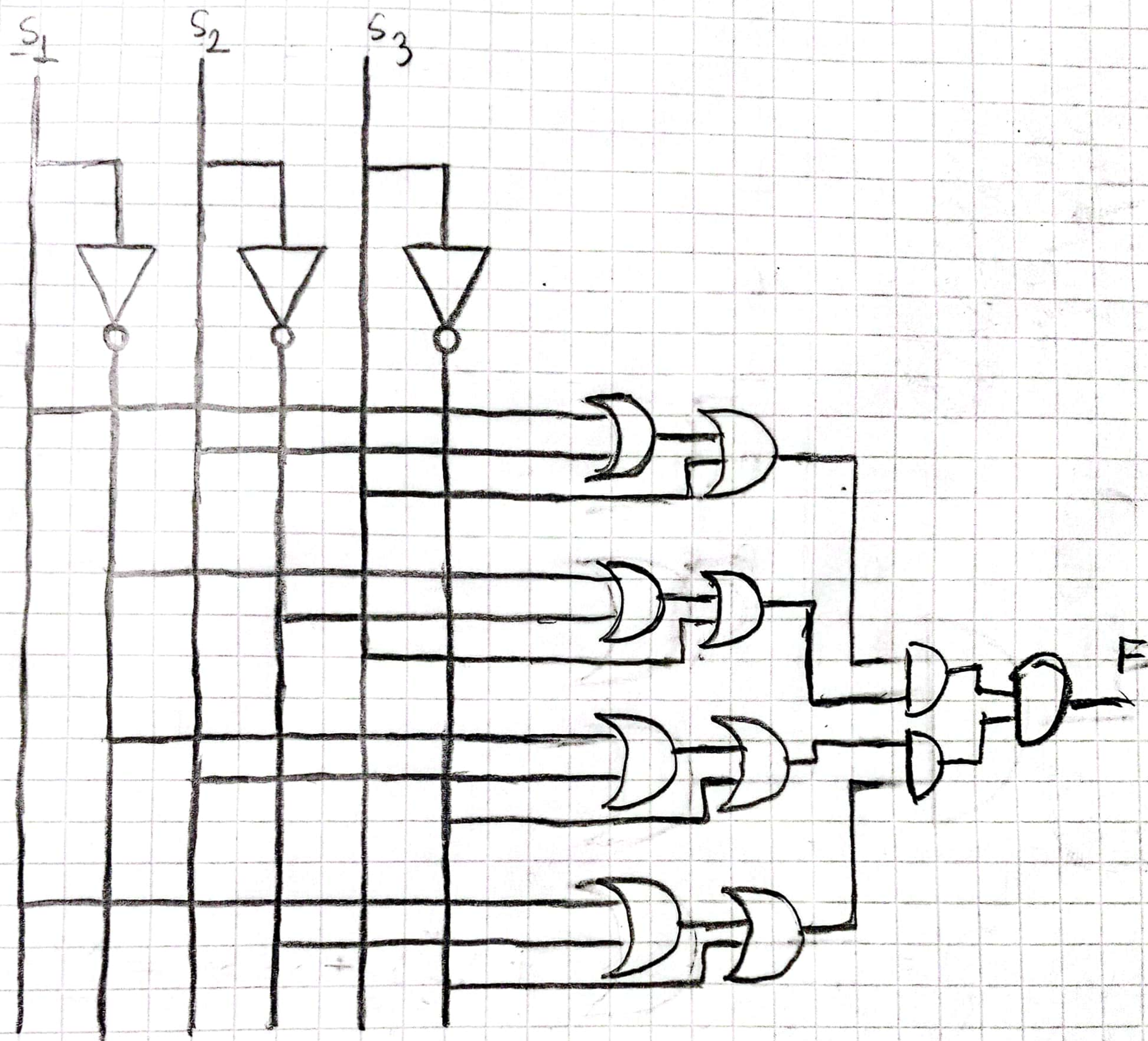
$$f(s_1, s_2, s_3) = \bar{S}_1 \bar{S}_2 S_3 + \bar{S}_1 S_2 \bar{S}_3 + S_1 \bar{S}_2 S_3 + S_1 S_2 S_3 = \sum(1,2,4,7)$$

3. Product-of-Sums expression:

$$f(s_1, s_2, s_3) = (S_1 + S_2 + S_3) . (S_1 + \bar{S}_2 + \bar{S}_3) . (\bar{S}_1 + S_2 + \bar{S}_3) . (\bar{S}_1 + \bar{S}_2 + S_3)$$

$$= \Pi(0,3,5,6)$$

4. logic Diagram:

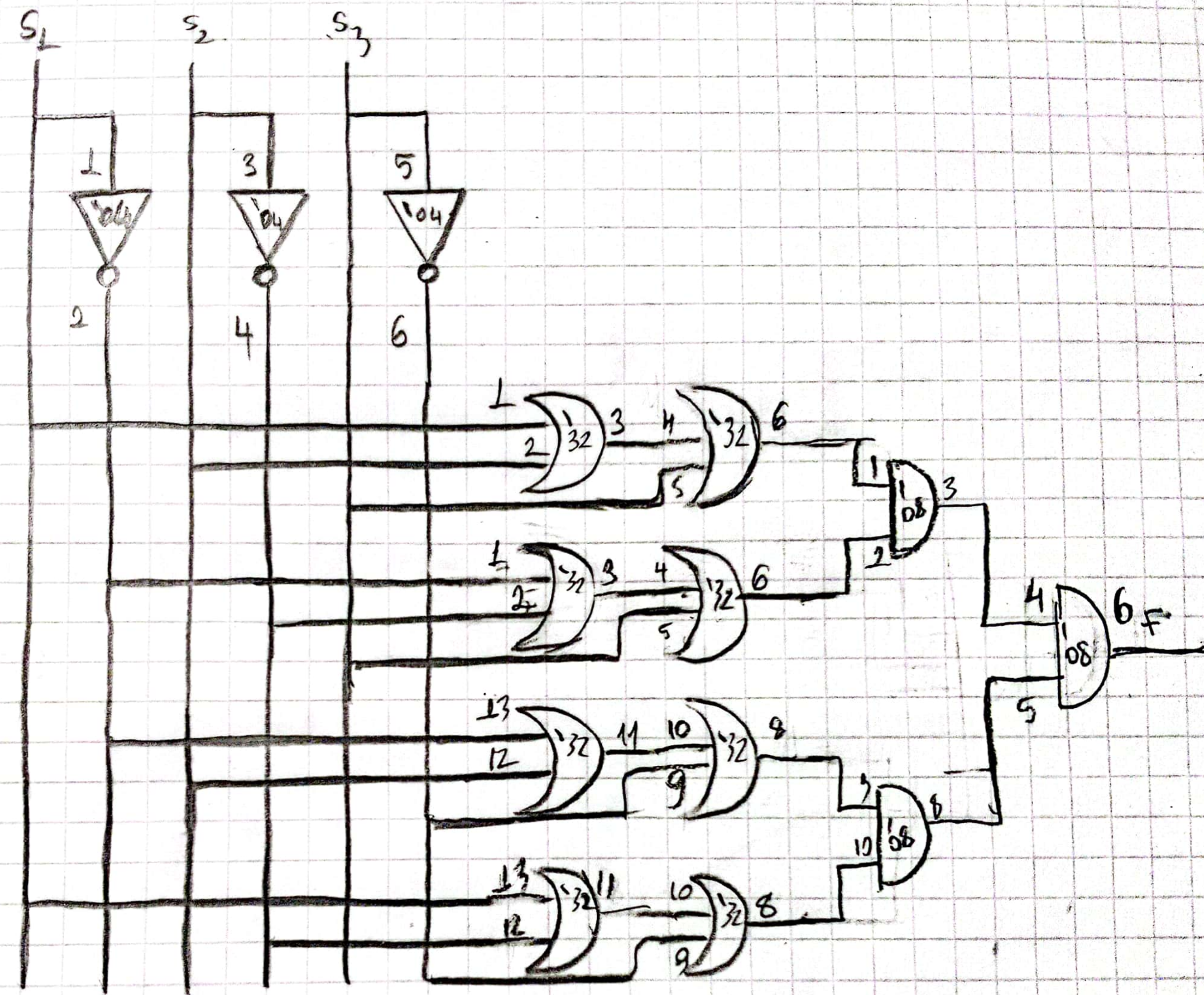


a) One 74LS08 Quad 2 input AND gate
b) Two 74LS32 Quad 2-input OR gate
c) One 74LS04 Hex INV gate

} Integrated Circuits

* In this logic diagram; 3 INV, 8 OR, 3 AND gates are planned to use.

6. Circuit Schematic:



- 74LS08 → GND - 7
  +5V - 14
- 74LS32 → GND - 7
  +5V - 14
- 74LS04 → GND - 7
  +5V - 14